DS5102: Big Data Lab

Lec 02

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The Problem

- In Big Data, “exactness” is expensive.
- Sorting 1 PB of data → Hours/Days.
- Checking if an element is present in a set of 1 billion items → GB’s of RAM

The Solution

- Randomized Algorithms
- We accept a tiny probability of error δ or a small approximation error ϵ?
- We will save on space and time!
- How can we trust the answer if it's "random"?

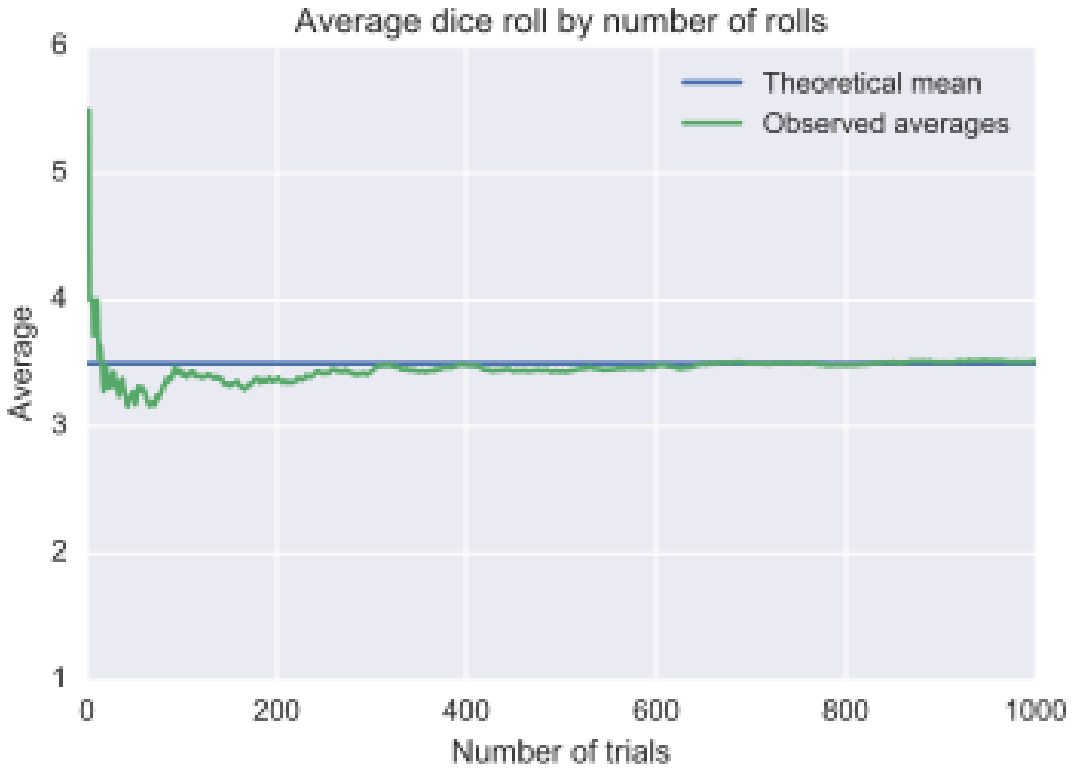
Law of Large Numbers (LLN)

X_i : i-th roll o/p

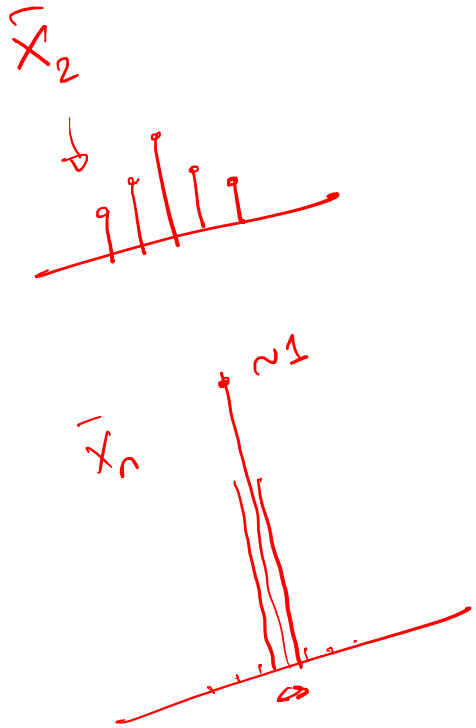
$X_i \in \{1, 2, 3, 4, 5, 6\}$ w.p. $1/6$ each

$$\mathbb{E}[X_i] = \frac{1}{6} \times 1 + \cdots\cdots + \frac{1}{6} \times 6$$

$$= \frac{21}{6} = \underline{\underline{3.5}}$$



Law of Large Numbers (LLN)



- Let $X_1, X_2, \ldots, X_n$ be i.i.d random variables
  - $E[X_i] = \mu$ and $var(X_i) = \sigma^2 < \infty$
- Define the sample mean

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \,.$$

- As $n \to \infty$, the sample mean converges in probability to the true mean μ.

Chebyshev's Inequality

- Let X be a random variable with mean μ and finite variance σ^2. For any $\epsilon > 0$, we have:

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

- **Intuition**: For any random variable with small variance, probability of taking values far from the mean is very small.

$X_1, X_2, \cdots\cdots, X_n$ i.i.d. mean: μ & var: σ^2

$$\bar{X}_n = \left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)$$

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n}\left[\mathbb{E}(X_1) + \cdots\cdots + \mathbb{E}(X_n)\right]$$

$$= \frac{n\mu}{n} = \mu$$

$$\mathrm{var}(\bar{X}_n) = \frac{1}{n^2}\left[\mathrm{var}(X_1) + \cdots + \mathrm{var}(X_n)\right]$$

$$= \frac{n\sigma^2}{n^2} = \left(\frac{\sigma^2}{n}\right)$$

$$P_r\left[|\bar{X}_n - \mu| \geq \epsilon\right] \leq \frac{\sigma^2}{n\epsilon^2} \longrightarrow 0 \quad \text{as } n \uparrow$$

$\frac{1}{n}$ decay

$\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y)$

if X & Y are independent

$\mathrm{var}(aX) = a^2\,\mathrm{var}(X)$

Hoeffding's Inequality

$X_i \Rightarrow \{0, 1\}$

- Let $X_1, X_2, \ldots, X_n$ be i.i.d Bernoulli random variables with mean μ

- $\boxed{P\left(\left|\overline{X}_n - \mu\right| \geq \epsilon\right) \leq 2e^{-2n\epsilon^2}}$

- *Chebyshev:* "You are unlikely to be wrong.“

- *Hoeffding:* "You are astronomically unlikely to be wrong."

Example 1: Random Sampling

- We are interested in conducting an election survey on a population of 1 billion.

- Each person will vote for either party A or B. Each person already made up his/her mind and there is no probability/randomness here.

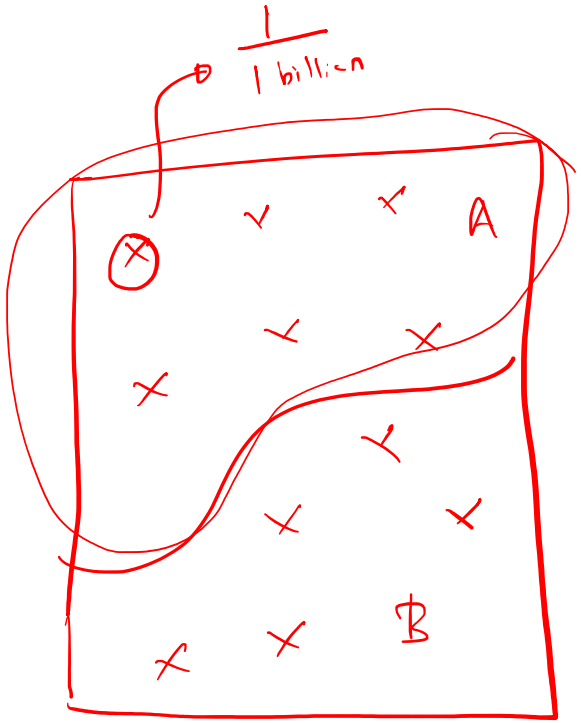
- We are interested in knowing the total votes or the percentage of votes party A will get.

→ Suppose true vote share for A : p

→ X_i : You pick a person uniformly at random and ask whom she voted for

$X_i \to 0$: if they voted for person B

1 : " person A



→ "Algo" : You sample uniformly at random (with replacement) n times

↳ $X_1, X_2, \ldots\ldots, X_n$

$$P[X_i = 1] = \frac{\#\text{people who voted for A}}{1 \text{ billion}} = p$$

$$P[X_i = 0] = 1-p$$

$$\mathbb{E}[X_i] = 1 \times p + 0 \times (\ ) = p$$

$$P\left[\,|\bar{X}_n - p| \geq \epsilon\,\right] \leq 2e^{-2n\epsilon^2}$$

$$P\left[\,|\bar{X}_n - p| \geq 0.01\,\right] \leq \boxed{2e^{-2n(0.01)^2} \leq 0.02}$$

$$-2n(0.01)^2 \leq \ln(0.01)$$

$$\Rightarrow n \geq \left\lceil \frac{4.61}{2\epsilon^2} \right\rceil = \underline{\underline{23026}}$$

$\epsilon \to 0.01$

"Your estimate is within 0.01 of the true vote share p with a confidence of $\geq 98\%$"

Example 2: Bloom Filter

Ref: Probability & Computing
Mitzenmacher & Upfal
[Chap. 5]

- You are building a malicious URL detector
- You have a blacklist of 1 billion dangerous URLs
- Store them in a set
- Average URL length ~100 bytes

Storing → 100×10^9 bytes $\approx$ (100 GB)

Bloom filter → 10×10^9 bits ≈ 1 GB

0: False negative (you are not going to miss out on dangerous URLs)

0.01 : False positive (you may classify some safe URLs as unsafe by mistake)

→ $S = \{s_1, s_2, \ldots, s_m\} \subseteq \mathcal{U}$ list of m items from a large universe $\mathcal{U}$

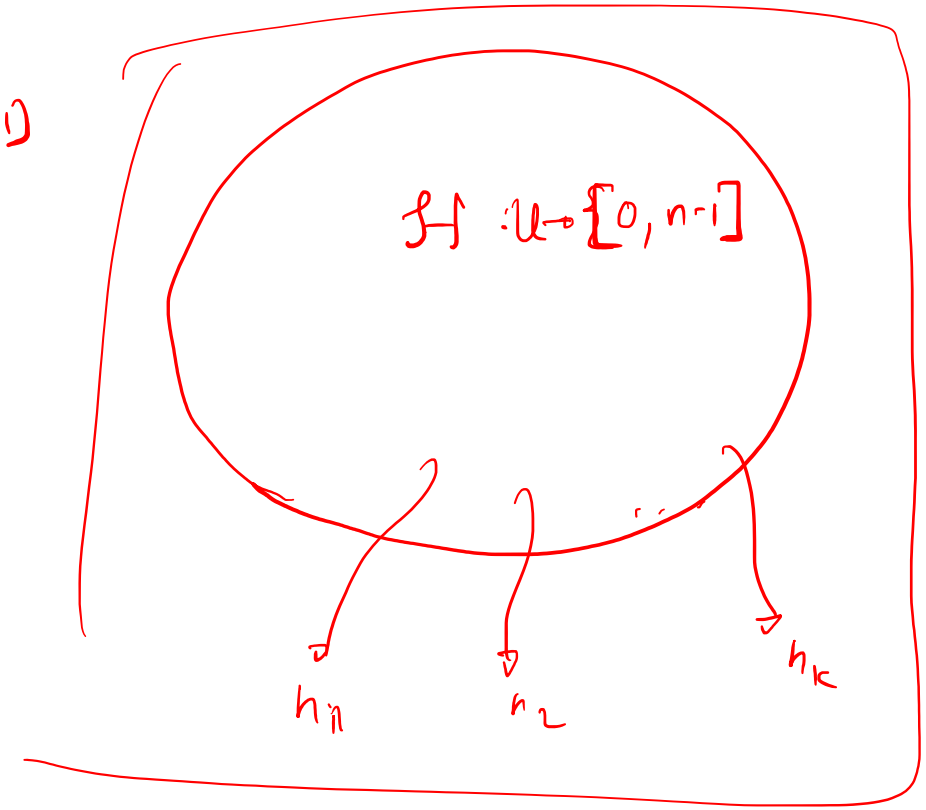
→ data structure with n bits (Bloomfilter)

A $\qquad$ $A[0], \ldots, A[n-1]$

↑

initialized with all-zero

→ random hash functions $h_1, h_2, \ldots h_k : \mathcal{U} \to [0, n-1]$



Pick some $u \in \mathcal{U}$

Fix any $i \in [0, n-1)$ $\qquad$ $\Pr[h_i[u] = i] = \frac{1}{n}$

→ For each $s_i \in S$

Apply the hash functions $h_1(\cdot), \ldots, h_k(\cdot)$ on s_i

Set $A[h_j[s_i]] = 1$ for $j = 1, 2, \ldots, k$

→ m - items ; n - bits in the array ; k - hash functions

What is the prob. that a specific bit of the array is still 0 after the algorithm saw & went through m items
(say $i \in [0:n-1]$)

$$\approx \left(1 - \frac{1}{n}\right)^{km} \Rightarrow e^{-\frac{km}{n}}$$ (for very large n)

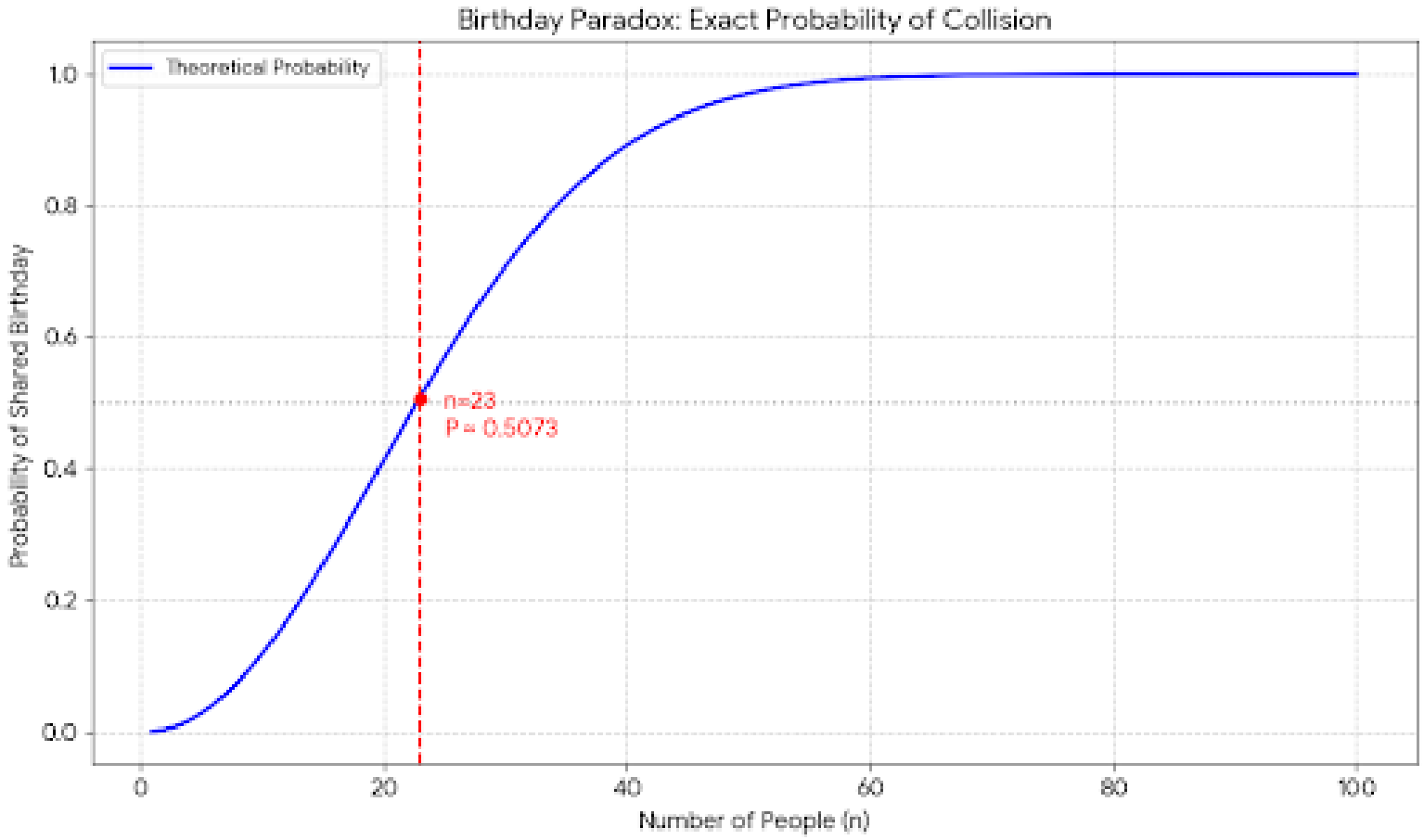
0 w.p. $\approx \underbrace{e^{-\frac{km}{n}}}_{p}$

0 w.p. $\approx \underbrace{e^{-\frac{km}{n}}}_{p}$

→ when $n \uparrow$, the fraction of 0's within the array $\approx np$

→ False positive prob. $\approx \dfrac{((1-p)n)^k}{n^k} = (1-p)^k \rightarrow 0.01$

$k \approx 7$; $n \approx 10m$


Birthday Paradox: Exact Probability of Collision
Theoretical Probability
n=23
P ≈ 0.5073
Probability of Shared Birthday
Number of People (n)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100

Thank You!